



Assessment of Urban Sprawl and Municipal Planning Challenges in the Kaduwela Municipal Council Area, Sri Lanka

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Abstract: Urbanization in rapidly expanding municipalities in developing countries presents significant spatial and governance challenges. This study examines urban sprawl and related municipal planning issues in the Kaduwela Municipal Council (KMC) area of Sri Lanka, using land-use and land-cover (LULC) data for 2002, 2012, and 2024 and demographic data up to 2023. By integrating Geographic Information System (GIS)-based land-use analysis with regression modeling, the study investigates land-use transformation, population change, and selected factors associated with urban expansion. Landsat satellite imagery from 2002, 2012, and 2024 was used to classify major land-use and land-cover categories, including built-up areas, vegetation, agricultural land, water bodies, and bare land. The classified maps were used to construct an Urban Sprawl Index (USI) to assess the extent and pattern of urban sprawl over the study period. Regression analysis was then applied to examine the relationship between the USI and selected demographic, infrastructural, and socioeconomic variables, including population growth, population density, road density, vehicle density, employment rate, and sectoral population distribution. The results indicate substantial land-use transformation in KMC, with an expansion of built-up areas and a decline in agricultural land, vegetation, and water bodies. The regression results show that population growth rate was the only statistically significant predictor of the USI, while other variables showed weak or non-significant associations. These findings suggest that urban sprawl in KMC is shaped by both measurable demographic factors and other contextual factors, such as land-use regulation, environmental constraints, informal development, and municipal governance capacity. The study highlights the need for integrated land-use planning, improved GIS-based monitoring, stronger zoning enforcement, infrastructure coordination, and environmental protection to support more sustainable urban management in Kaduwela and similar peri-urban municipalities in Sri Lanka.

Keywords: Urban sprawl; GIS-based spatial analysis; Municipal governance; Land-use change; Sustainable urban development; Urban Sprawl Index

1 Introduction

Urbanization is one of the major forces shaping contemporary cities, economies, and environments. The rapid expansion of urban areas has created opportunities for economic development and improved access to services, but it has also intensified challenges related to land-use change, infrastructure provision, environmental degradation, and municipal governance. These challenges are particularly significant in developing countries, where urban growth often occurs faster than planning institutions and infrastructure systems can respond [1, 2].

Urban sprawl is a common outcome of poorly managed urbanization. It is generally associated with fragmented land development, conversion of agricultural and natural areas, inefficient infrastructure expansion, and increasing pressure on public services [3, 4]. For municipalities, managing urban sprawl requires not only spatial planning tools but also effective governance, reliable data, and coordinated policy implementation. Without such measures, urban expansion may lead to long-term environmental, social, and economic costs.

Sri Lanka has experienced increasing urbanization in recent decades, especially within and around the Colombo Metropolitan Region [5, 6]. As urban growth extends beyond the central city, peri-urban municipalities have become important zones of transformation. These areas often experience rapid changes in land use, population distribution,

infrastructure demand, and environmental conditions. However, compared with major urban centers, peri-urban municipalities have received relatively limited research attention.

The Kaduwela Municipal Council (KMC), located in the Colombo District of Sri Lanka's Western Province, provides a relevant case for examining these issues. The municipality has gradually transformed from a predominantly rural and semi-urban area into a rapidly urbanizing suburban corridor connected to the wider Colombo metropolitan area. This transformation has created development opportunities, but it has also increased pressure on land, infrastructure, drainage systems, green spaces, and municipal service delivery.

Against this background, this study examines urban sprawl and related municipal challenges in the KMC area from 2002 to 2024. The study combines Geographic Information System (GIS)-based land-use analysis with quantitative regression modeling to assess spatial change and selected drivers of urban expansion. By linking land-use transformation, demographic change, and municipal governance concerns, the study aims to provide evidence that can support more sustainable planning and management in Kaduwela and similar peri-urban municipalities in Sri Lanka.

This study contributes to the literature in three main ways. First, it provides a case-specific analysis of urban sprawl in a rapidly changing peri-urban municipality in Sri Lanka. Second, it applies GIS-based spatial analysis and regression modeling to examine changes in land use and selected factors associated with urban sprawl. Third, it discusses the planning and governance implications of these changes, with particular attention to land-use management, infrastructure provision, environmental protection, and municipal coordination.

1.1 Research Problem

Rapid urbanization in peri-urban municipalities creates complex challenges for local planning and governance. In the KMC area, population growth, infrastructure expansion, and increasing demand for residential, commercial, and institutional land have contributed to significant spatial transformation. These changes have placed pressure on municipal services, transport systems, drainage networks, agricultural land, and environmentally sensitive areas.

Although Kaduwela plays an important role in the expansion of the Colombo Metropolitan Region, the municipality faces practical constraints in managing urban growth. Unplanned or weakly regulated development can lead to fragmented land conversion, loss of green and agricultural areas, increased flood risk, and uneven access to infrastructure and services. These issues highlight the need for a clearer understanding of how urban sprawl has developed in the area and what factors are associated with this process.

The core research problem of this study is therefore to assess how urban sprawl has emerged in the KMC area and how it affects municipal planning and management. Specifically, the study examines spatial and demographic changes over time, identifies selected factors associated with the Urban Sprawl Index (USI), and discusses planning strategies that may help strengthen sustainable urban governance in Kaduwela.

1.2 Research Objectives

The main objective of this study is to assess urban sprawl and related municipal challenges in the KMC area and to propose planning and governance strategies for more sustainable urban development.

The specific objectives are as follows:

1. To analyze population change and land-use transformation in the KMC area using demographic data up to 2023 and land-use and land-cover (LULC) data for 2002, 2012, and 2024.
2. To examine the spatial patterns of urban expansion using GIS-based land-use and land-cover analysis.
3. To assess urban sprawl through the USI and examine its relationship with selected demographic, infrastructural, and socioeconomic variables.
4. To identify key municipal planning and governance challenges associated with urban sprawl in the KMC area.
5. To propose context-appropriate recommendations for sustainable land-use management, infrastructure planning, and urban governance.

1.3 Research Questions

This study is guided by the following research questions:

1. How have population distribution and land-use/land-cover patterns changed in the KMC area based on demographic data up to 2023 and LULC data for 2002, 2012, and 2024?
2. What spatial patterns of urban expansion and sprawl can be identified through GIS-based analysis and the USI?
3. How is the USI related to selected demographic, infrastructural, and socioeconomic variables?
4. What planning and governance strategies can support more sustainable urban growth in the KMC area?

1.4 Significance of the Research

This study has both academic and practical significance. Academically, it contributes to the understanding of urban sprawl in peri-urban municipalities, particularly in the context of Sri Lanka. While many studies focus on

major metropolitan centers, this study highlights the importance of examining municipalities located at the edge of metropolitan expansion, where land-use change and governance challenges are often highly visible.

Methodologically, the study combines GIS-based spatial analysis with regression modeling to examine urban sprawl from both spatial and quantitative perspectives. This approach helps link observable land-use changes with selected demographic and infrastructural factors, providing a more structured basis for understanding urban expansion in Kaduwela.

Practically, the findings can support municipal decision-making by identifying areas of land-use pressure, infrastructure demand, and environmental concern. The study may assist planners and policymakers in developing more targeted strategies for land-use regulation, infrastructure planning, flood-risk management, and protection of remaining green and agricultural areas. The findings may also provide useful lessons for other rapidly urbanizing municipalities in Sri Lanka.

1.5 Research Limitations

This study has several limitations. First, the analysis depends mainly on available secondary data, satellite imagery, and administrative records. The accuracy and completeness of these data may affect the precision of the findings.

Second, land-use classification based on satellite imagery may involve classification errors, especially in areas where land-cover categories are visually similar or spatially mixed. Although classification procedures can reduce such errors, some uncertainty remains.

Third, the regression analysis includes selected demographic, infrastructural, and socioeconomic variables, but urban sprawl is also influenced by factors that are difficult to quantify, such as land-market behavior, policy enforcement, informal development, and local political decisions. Therefore, the model may not fully capture all drivers of urban expansion.

Fourth, the study focuses only on the KMC area. While the findings may be useful for understanding similar peri-urban municipalities, they should not be generalized without considering local differences in geography, governance capacity, infrastructure, and development pressure.

Finally, broader factors such as climate change, economic fluctuations, and national policy changes may also influence urban development in Kaduwela. These factors are acknowledged but are not examined in detail within the scope of this study.

2 Literature Review

2.1 Conceptualizing Urban Sprawl and Emerging Urbanization Trends

Urban sprawl generally refers to the unplanned or poorly regulated outward expansion of urban areas into rural and peri-urban land. It is commonly associated with low-density development, fragmented land conversion, automobile dependence, and inefficient use of infrastructure and public services [3]. Although urban sprawl was once discussed mainly in relation to industrialized countries, it has become increasingly visible in developing regions, where rapid population growth, infrastructure expansion, and weak land-use regulation often accelerate peri-urban transformation [1, 7].

The consequences of sprawl are multidimensional. Environmentally, it may contribute to the loss of agricultural land, wetlands, vegetation cover, and biodiversity. Socially and economically, it can increase infrastructure costs, widen service inequalities, and create difficulties for municipal planning and governance. These impacts are particularly significant in developing-country municipalities, where institutional capacity, technical resources, and financial support are often limited.

Recent studies have used spatial tools such as remote sensing, Shannon's entropy, urban expansion indices, and GIS-based land-use analysis to measure and monitor sprawl patterns [8]. These tools are useful for identifying the direction, intensity, and distribution of urban expansion. However, understanding urban sprawl also requires attention to local governance, planning practices, infrastructure provision, and environmental constraints.

2.2 Theoretical Foundations of Urban Growth and Sprawl

Theories of urban growth provide different explanations for why cities expand and how land-use patterns change over time. Classical urban models, such as the concentric zone model and bid-rent theory, explain urban expansion through accessibility, land value, and distance from the urban center. These models remain useful, but they are often insufficient for explaining the complex and fragmented growth patterns observed in contemporary peri-urban areas.

More recent perspectives emphasize the interaction between spatial, institutional, economic, and environmental factors. Spatial theories highlight the role of transport networks, land markets, and residential preferences in shaping suburbanization and dispersed growth. Institutional perspectives focus on the influence of planning regulations, municipal capacity, and land-use governance [2]. Ecological and sustainability-oriented perspectives further stress the need to balance urban growth with environmental protection, green infrastructure, and climate resilience [9].

Complexity theory also provides a useful lens for understanding urban sprawl. It views cities as adaptive systems in which land-use change emerges from the interaction of multiple actors, policies, markets, and environmental conditions [10]. This perspective is relevant to rapidly urbanizing municipalities such as Kaduwela, where demographic growth, infrastructure development, local land-use decisions, and environmental constraints interact to shape spatial expansion.

2.3 Determinants and Indicators of Urban Sprawl

Urban sprawl is influenced by a range of demographic, infrastructural, socioeconomic, and institutional factors. Population growth is one of the most frequently examined drivers, as increasing population pressure often raises demand for housing, services, and developable land [4, 11]. However, the relationship between population growth and sprawl is not always direct. In some contexts, growth may lead to outward expansion, while in others it may be absorbed through densification or infill development [11].

Infrastructure and mobility factors also play an important role. Road density, transport accessibility, and vehicle ownership can influence the location and pattern of new development. In areas where transport systems are poorly integrated with land-use planning, infrastructure expansion may encourage fragmented or linear development along major corridors [2, 9].

Socioeconomic factors, including income levels, employment structure, housing demand, and land-market dynamics, can further shape urban expansion. Weak planning control, speculative land development, and informal settlement growth may intensify unregulated land conversion, particularly in peri-urban municipalities with limited enforcement capacity [12].

Based on these considerations, this study examines selected demographic, infrastructural, and socioeconomic variables in relation to urban sprawl in the KMC area. These include population growth, population density, in-migration, household density, road density, vehicle density, employment rate, and sectoral population distribution. These variables are used to explore how measurable local factors are associated with the USI. However, because urban sprawl is also shaped by institutional, environmental, and informal land-use processes, the interpretation of statistical results should remain cautious.

2.4 Municipal Governance Challenges in Managing Urban Growth

Municipalities play a central role in managing urban growth, but they often face significant constraints in responding to rapid expansion. Common challenges include limited financial resources, weak land-use enforcement, fragmented institutional responsibilities, inadequate infrastructure, and insufficient technical capacity [12]. These constraints can make it difficult for local authorities to guide development in a coordinated and sustainable manner.

Urban sprawl increases the cost and complexity of municipal service delivery. Dispersed development requires wider extension of roads, drainage, water supply, waste management, and public transport services. It may also create spatial inequalities when some areas receive better services than others. In environmentally sensitive areas, such as floodplains and wetland zones, unregulated development can also increase disaster risk and reduce ecological resilience.

For peri-urban municipalities, governance challenges are particularly complex because these areas often experience rapid land conversion before planning systems are fully prepared. Effective management therefore requires updated zoning, stronger enforcement, inter-agency coordination, GIS-based monitoring, and integration between land-use planning, infrastructure planning, and environmental protection.

2.5 Geographic Information System and Quantitative Tools for Assessing Urban Sprawl

GIS and remote sensing have become important tools for assessing urban sprawl because they allow spatially explicit analysis of land-use and land-cover change. Landsat imagery and other satellite data can be used to detect changes in built-up areas, vegetation, agricultural land, bare land, and water bodies over time [13]. These methods are especially useful in contexts where field-based land-use records are limited or inconsistent.

Several quantitative indicators have been used to measure sprawl, including Shannon's entropy, urban expansion indices, landscape metrics, and built-up density measures [3]. In addition, regression analysis can be used to examine the relationship between urban sprawl indicators and possible explanatory variables, such as population growth, infrastructure access, and socioeconomic characteristics.

More advanced models, including Markov chains, cellular automata, artificial neural networks, and scenario-based simulations, have also been applied to predict future land-use change [8]. While these methods provide useful insights, their reliability depends on the availability and quality of spatial data, model assumptions, and validation procedures. Therefore, studies using GIS and quantitative modeling should clearly explain data sources, classification methods, accuracy assessment, variable selection, and model diagnostics.

2.6 Empirical Studies on Urban Sprawl and Municipal Response

Empirical studies show that urban sprawl varies across regions depending on demographic pressure, infrastructure development, land-use policy, and governance capacity. Studies in China have linked urban expansion with land-use regulation, infrastructure development, and environmental impacts, including carbon emissions and energy use [14]. In India, studies of medium-sized cities have shown that remote sensing and GIS can effectively reveal the scale and pattern of urban expansion over time [13].

Research in Europe and North America has often focused on policy tools such as urban growth boundaries, compact development, and transit-oriented planning [3]. These studies suggest that effective regulation and coordinated infrastructure planning can reduce fragmented development and support more sustainable urban forms.

In the Sri Lankan context, research has examined urban growth and land-use transformation in the Colombo Metropolitan Region and other urbanizing areas [5, 6, 15]. However, more attention is still needed on peri-urban municipalities, where rapid growth, land conversion, environmental vulnerability, and limited municipal capacity intersect. Kaduwela is particularly relevant because it is closely connected to Colombo's metropolitan expansion while also containing environmentally sensitive areas, agricultural land, and rapidly changing residential and commercial zones.

2.7 Critical Appraisal and Research Gap

The existing literature provides important insights into the causes, patterns, and impacts of urban sprawl. However, several gaps remain. First, many studies focus on large metropolitan centers, while smaller and peri-urban municipalities receive less attention. This limits understanding of how urban sprawl affects local governance, service delivery, and environmental management in municipalities with more limited planning capacity.

Second, many studies examine spatial expansion, demographic change, or governance challenges separately. Fewer studies combine GIS-based land-use analysis with quantitative modeling and municipal governance interpretation in a single framework. Such integration is important because urban sprawl is both a spatial process and a governance challenge.

Third, in the Sri Lankan context, more localized studies are needed to examine how peri-urban municipalities within the Colombo Metropolitan Region are changing over time. Although previous studies have discussed urban growth in the broader Colombo region, there is still limited evidence on how land-use transformation, population change, and municipal planning challenges interact in the KMC area.

This study addresses these gaps by examining urban sprawl in KMC from 2002 to 2024 using GIS-based land-use analysis and regression modeling. By linking spatial change with selected demographic, infrastructural, and socioeconomic variables, the study provides case-specific evidence on urban expansion and its implications for municipal planning and governance in a rapidly urbanizing Sri Lankan municipality.

3 Methodology

3.1 Study Area

The KMC is located in the eastern part of the Colombo District in Sri Lanka's Western Province, as shown in Figure 1. It forms part of the wider Colombo Metropolitan Region and has experienced rapid land-use and demographic changes over recent decades. The municipality covers approximately 91.76 km² and consists of 57 Grama Niladhari (GN) divisions [16], as displayed in Figure 2.

For administrative and spatial analysis, the study area is considered through its major local divisions, including Kaduwela, Battaramulla, and Athurugiriya. These areas contain a mixture of urban, semi-urban, residential, commercial, institutional, agricultural, and environmentally sensitive land uses. The municipality is also characterized by wetlands, waterways, and low-lying areas associated with the Kelani River and Diyawanna Oya system, making flood risk and environmental management important planning concerns.

According to recent administrative estimates, the population of KMC was approximately 281,282 in 2023. The municipality has become an important suburban corridor connected to the expansion of Colombo, with growing residential, commercial, and government-related functions. At the same time, the conversion of agricultural and vegetated land, pressure on infrastructure, and environmental sensitivity of the area make KMC a relevant case for examining urban sprawl and municipal planning challenges.

3.2 Research Scope

This study focuses on urban sprawl and related municipal challenges in the KMC area from 2002 to 2024. The geographical scope is limited to the administrative boundary of KMC, including its 57 GN divisions. This allows the study to examine urbanization patterns at both the municipal and local GN-division levels.

The LULC analysis covers three study years: 2002, 2012, and 2024. The demographic analysis uses the closest available population data for 2002, 2012, and 2023. This temporal arrangement was adopted because the latest LULC

data and the latest population data were not available for exactly the same year. Therefore, the population-related results are interpreted as the closest available demographic baseline for the 2024 LULC-based analysis.

The unit of analysis includes the 57 GN divisions within KMC. This spatial unit allows comparison of population growth, land-use transformation, and urban sprawl patterns across different parts of the municipality. The findings are therefore intended to provide a local-level understanding of urban expansion and its implications for municipal planning and governance.

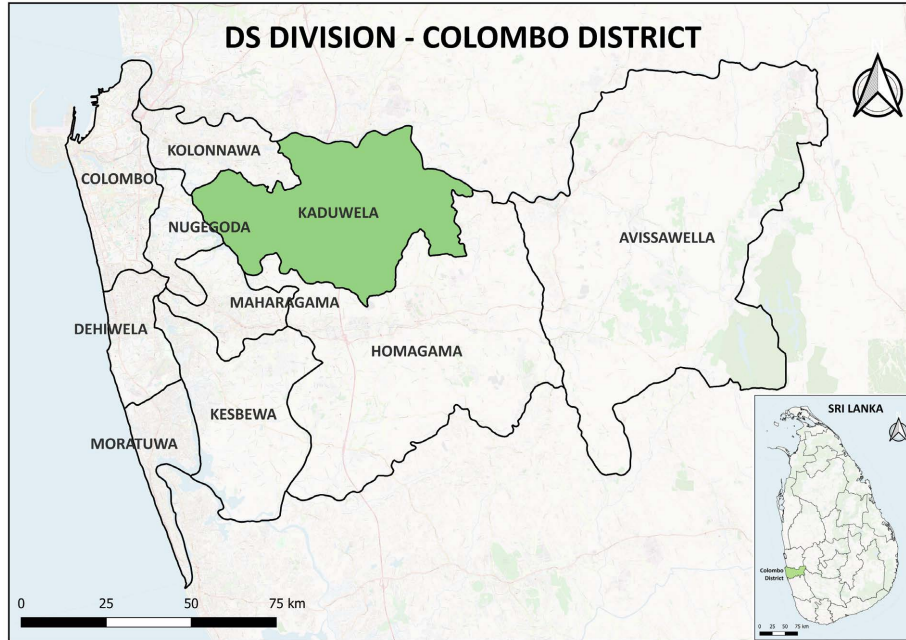


Figure 1. Location of Kaduwela Municipal Council (KMC) in Colombo District, Sri Lanka

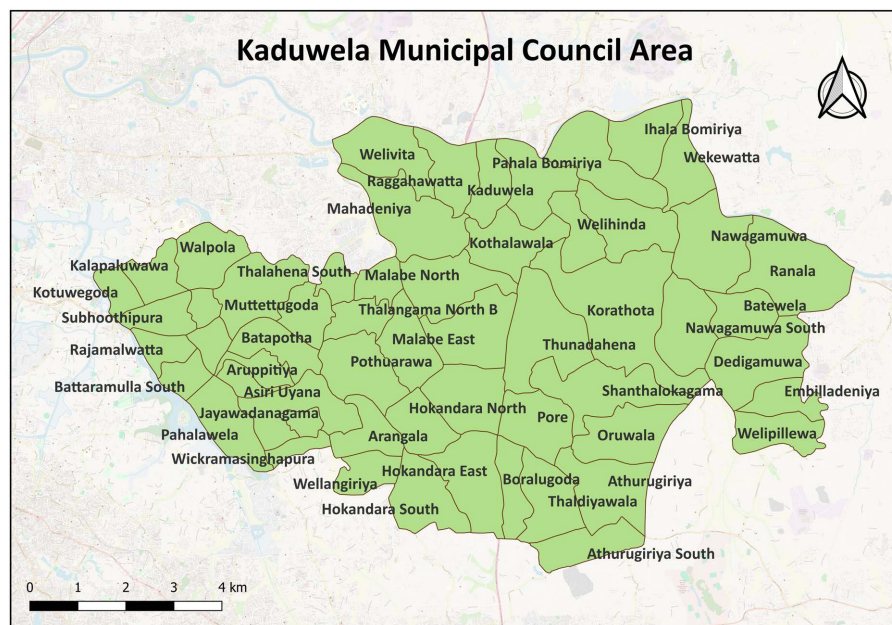


Figure 2. Grama Niladhari (GN) divisions in the Kaduwela Municipal Council (KMC) area

3.3 Research Approach

This study adopts a quantitative and spatial analytical approach. GIS-based land-use and land-cover analysis was used to identify spatial changes in the KMC area, while regression analysis was applied to examine the relationship

between the USI and selected explanatory variables. The combination of these methods allows the study to assess both the spatial pattern of urban expansion and the possible factors associated with it.

3.3.1 Research framework

The research framework is based on the relationship between land-use change, population dynamics, infrastructure conditions, and urban sprawl, as shown in Figure 3. Landsat satellite imagery was used to identify changes in major LULC categories, including built-up areas, agricultural land, vegetation, water bodies, and bare land. These spatial changes were then used to calculate the USI, which serves as the dependent variable in the regression analysis.

The independent variables were grouped into three categories. The first category includes population-based variables, such as population density, population growth rate, in-migration rate, and household density. The second category includes infrastructure and mobility variables, such as road density and vehicle density. The third category includes socioeconomic and sectoral variables, such as employment rate and urban or semi-urban population distribution.

This framework allows the study to examine how selected demographic, infrastructural, and socioeconomic factors are associated with urban sprawl in the KMC area.

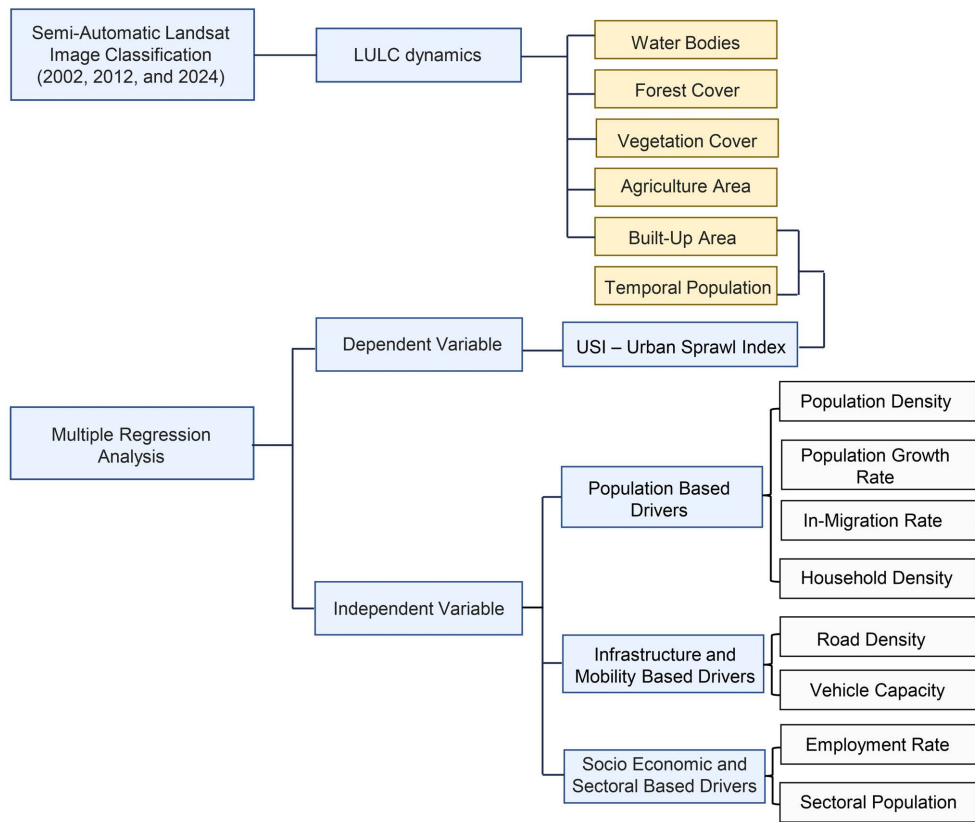


Figure 3. Research framework

3.3.2 Data collection, processing and interpretation

• Spatial-temporal analysis

Spatial-temporal analysis was conducted using GIS to examine the LULC changes in the KMC area. Landsat satellite images for 2002, 2012, and 2024 were obtained from the USGS EarthExplorer platform. The 2002 baseline was established using Landsat 7 Enhanced Thematic Mapper Plus (ETM+) imagery. For the 2012 baseline, Landsat 7 ETM+ imagery was initially used. Since Landsat 7 images acquired after 2003 may contain striping and data gaps due to the Scan Line Corrector failure (SLC-off), a gap-filling procedure was applied using a localized neighborhood-based focal mean approach and suitable cloud-free images from adjacent acquisition dates. Where necessary, the corrected 2012 dataset was cross-checked against available Landsat 5 Thematic Mapper (TM) imagery from the late 2011 to early 2012 period. Landsat 8 Operational Land Imager/Thermal Infrared Sensor (OLI/TIRS) imagery from 2013 was used only as supplementary reference data for visual validation and radiometric consistency checks, rather than as the primary data source for the 2012 classification. This multi-source validation approach was adopted to improve spatial continuity and enhance the reliability of the final LULC classification.

The satellite images were preprocessed before classification. The preprocessing steps included atmospheric correction, geometric correction, reprojection to UTM Zone 44N using WGS 84, cloud masking, and clipping to the KMC boundary. The images were then classified into four LULC categories: built-up area, agricultural land, vegetation cover, and water bodies.

A supervised classification approach using the Maximum Likelihood Classifier (MLC) was applied in QGIS. Classification accuracy was assessed using reference data and high-resolution Google Earth imagery. A confusion matrix, overall accuracy, and Kappa coefficient were used to evaluate classification reliability.

Post-classification comparison was used to identify changes in LULC categories across 2002, 2012, and 2024. The resulting maps and area statistics were used to examine the direction and scale of land-use transformation in the KMC area.

- **Urban Sprawl Index calculation**

The USI was used to quantify the extent and intensity of urban sprawl in the KMC area. The USI was calculated for each GN division based on the ratio of the land consumption rate (LCR) to the population growth rate (PGR), following the logic of SDG Indicator 11.3.1, which measures land-use efficiency through the ratio of land consumption rate to population growth rate [17]. The index was calculated as follows:

$$USI_i = \frac{LCR_i}{PGR_i} \quad (1)$$

where, USI_i is the USI for GN division i ; LCR_i is the Land Consumption Rate, which measures the rate of built-up area expansion in GN division i ; and PGR_i is the Population Growth Rate, which measures the rate of local population growth in GN division i .

Because GN-level population data were available up to 2023, PGR_i was calculated using the closest available demographic period.

- **Regression analysis**

Multiple regression analysis was used to examine the relationship between the USI and selected explanatory variables. The USI was used as the dependent variable. The independent variables included population density, population growth rate, in-migration rate, household density, road density, vehicle density, employment rate, and sectoral population distribution.

The regression model can be expressed as follows:

$$USI_i = \beta_0 + \beta_1 PD_i + \beta_2 PGR_i + \beta_3 IM_i + \beta_4 HD_i + \beta_5 RD_i + \beta_6 VD_i + \beta_7 ER_i + \beta_8 USP_i + \beta_9 SSP_i + \varepsilon_i \quad (2)$$

where, PD is population density; IM is in-migration rate; HD is household density; RD is road density; VD is vehicle density; ER is employment rate; USP is urban sector population; SSP is semiurban sector population; β_0 is the intercept; β_1 to β_9 are regression coefficients; and ε_i is the error term. Accordingly, population-related explanatory variables were interpreted as the closest available demographic baseline.

Model performance was assessed using R^2 , adjusted R^2 , p -values, and the F -statistic.

- **Data interpretation**

The results from the GIS-based LULC analysis and regression analysis were interpreted together to understand the spatial and statistical dimensions of urban sprawl in KMC. The LULC analysis was used to identify where and how land-use changes occurred, while the regression analysis was used to examine which selected variables were statistically associated with the USI.

The interpretation focused on three aspects: changes in land-use patterns, demographic and infrastructural factors related to urban sprawl, and the implications of these changes for municipal planning and governance. Since urban sprawl is influenced by both measurable and non-measurable factors, the regression results were interpreted cautiously, especially where variables were not statistically significant.

4 Results and Discussion

4.1 Population and Population Growth, 2002, 2012, and 2023

The population and growth rate analysis of the KMC area reveals significant demographic transitions between 2002 and 2023, offering critical insights into the underlying drivers of emerging urbanization and sprawl. By assessing population distribution and growth dynamics at the GN division level, the analysis identifies distinct spatial variations that are relevant to municipal planning and governance challenges.

The total population of KMC increased consistently over the two decades, from 209,251 in 2002 to 252,041 in 2012, and further to 281,282 in 2023 [16]. Spatial mapping indicates a notable densification in the central and western GN divisions, where darker shades on the maps represent higher population concentrations in 2012 and 2023 compared to 2002 (Figure 4). Peripheral areas, while initially less dense, have also shown signs of population expansion, particularly in the eastern and southern zones, reflecting gradual peri-urbanization.

These demographic shifts demonstrate that population pressures are spatially uneven. Central zones absorbed much of the population increase, reinforcing urban concentration, while outlying GN divisions exhibited emerging densification trends.

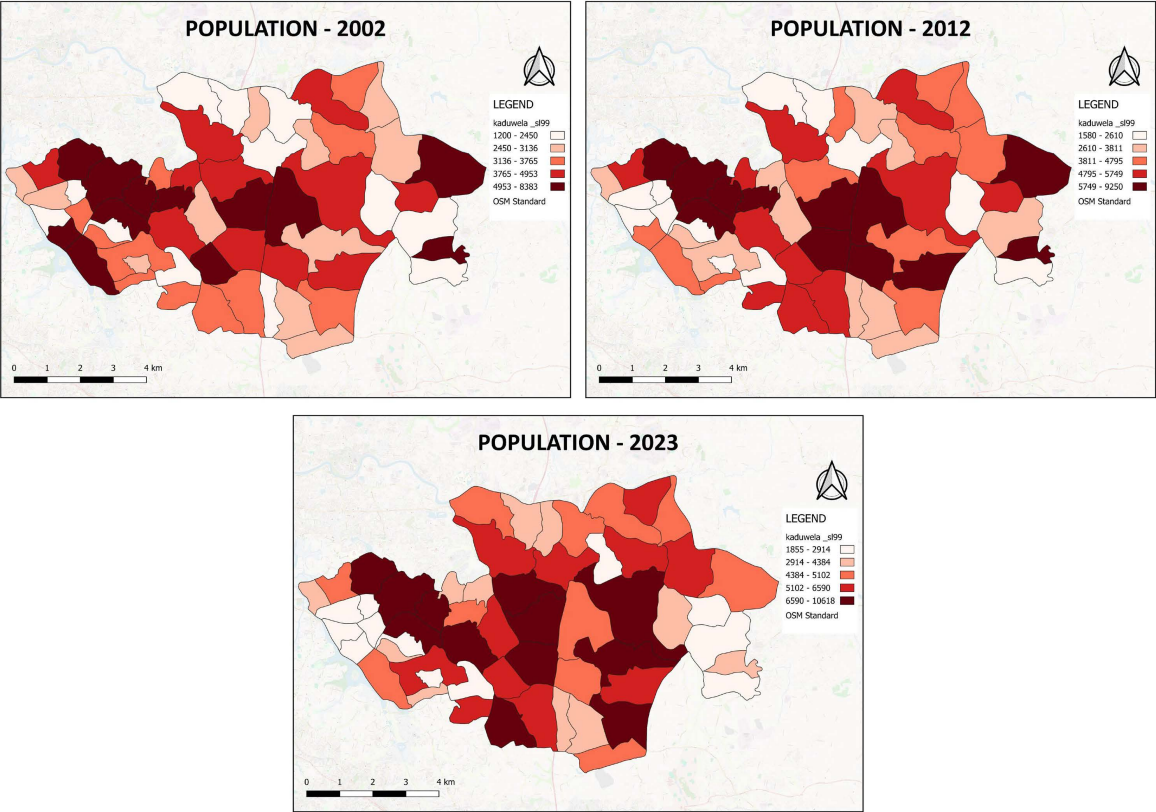


Figure 4. Population distribution in the Kaduwela Municipal Council (KMC) area in 2002, 2012, and 2023

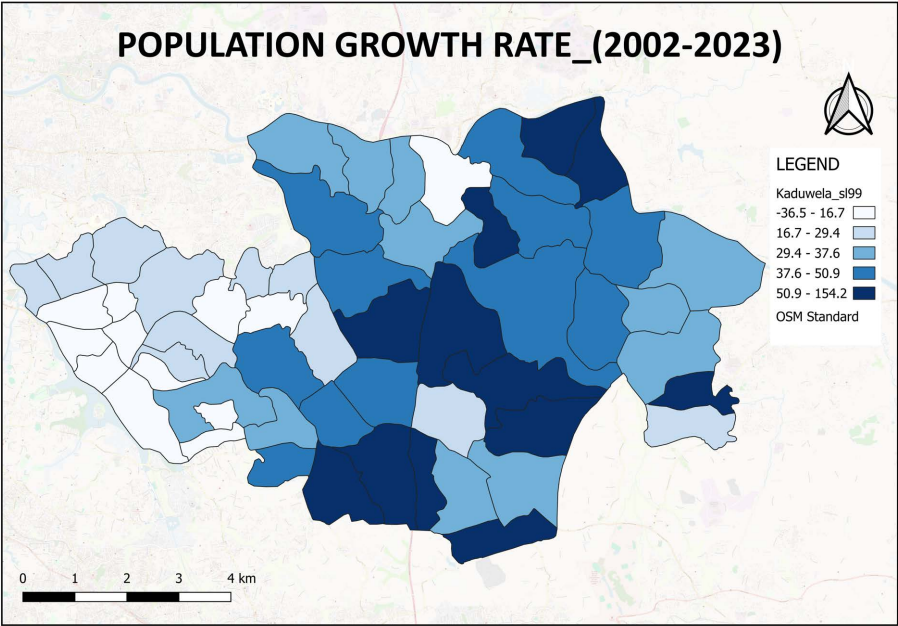


Figure 5. Population growth rate in the Kaduwela Municipal Council (KMC) area, 2002–2023

The spatial analysis of population growth rates further highlights the heterogeneity of urban development across KMC (Figure 5). Several GN divisions in the northern and central sectors recorded growth rates exceeding 200%,

suggesting rapid population concentration in selected areas. Conversely, a few divisions in the southern and peripheral areas exhibited low or even negative growth, suggesting selective migration patterns or uneven local development conditions.

Two key implications emerge from this demographic analysis. First, the sustained population increase exerts mounting pressure on land, housing, infrastructure, and essential services, thereby intensifying the risk of unplanned urban sprawl. Second, the uneven spatial distribution of growth across GN divisions reveals KMC's dual challenge: managing densely populated urban cores while simultaneously addressing the expansion of semi-urban and peri-urban zones.

These findings emphasize the need for spatially responsive planning and differentiated municipal policies that align infrastructure investment, land-use regulation, and service delivery with emerging demographic patterns. Such interventions are essential for mitigating uncontrolled urban expansion and promoting sustainable urban growth, which should be further examined together with land-use and regression results in the following sections.

4.2 Land Use Change Detection Using QGIS, 2002–2024

The LULC change detection analysis of the KMC area between 2002 and 2024, based on GIS analysis, reveals transformations across major land categories, including built-up areas, forest cover, agricultural land, and water bodies. These categories were mapped and compared across three timeframes: 2002, 2012, and 2024. The overall spatial pattern of LULC transformation is shown in Figure 6, while the percentage distribution of major LULC categories is summarized in Figure 7.

As shown in Figure 7, built-up land increased substantially from approximately 18% in 2002 to 33% in 2012 and 52% in 2024. In contrast, water bodies declined from approximately 18% to 6%, while vegetation cover decreased from approximately 33% to 17% during the same period. Agricultural land also declined overall, although it remained one of the major land-use categories in the KMC area. These changes indicate a clear shift from agricultural and vegetated landscapes toward increasingly built-up urban environments. Similar patterns of urban expansion and the conversion of vegetated or rural land have been reported in previous Colombo-based and Kaduwela-focused studies [5, 6, 15].

As shown in Figure 8, water bodies exhibited a gradual but notable decline between 2002 and 2024. In 2002, Kaduwela contained numerous ponds, wetlands, and small streams that supported ecological balance and groundwater recharge. By 2012, many smaller water bodies had disappeared, often reclaimed for residential or infrastructure development. The 2024 analysis reveals a continued contraction, with only a few prominent water bodies remaining. This decline highlights significant environmental stress, particularly in terms of drainage and flood regulation. Given that Kaduwela lies within a flood-prone basin, the loss of wetlands and surface water has aggravated flood risks, reducing the municipality's natural capacity for stormwater retention and management.

Similarly, vegetation cover declined steadily throughout the study period (Figure 9). In 2002, forested and green areas were relatively more extensive, especially in the northern and eastern regions of Kaduwela. However, by 2012, these areas were increasingly cleared for housing and mixed-use developments. By 2024, forest cover had become highly fragmented and disconnected, resulting in ecological degradation and loss of biodiversity. This fragmentation diminishes ecological resilience and contributes to climate-related challenges such as increased flooding, higher surface temperatures, and declining air quality. The reduction of vegetation also weakens the ecosystem's ability to provide critical services like carbon sequestration, erosion control, and temperature regulation, emphasizing the need for targeted reforestation and urban greening initiatives.

As displayed in Figure 10, agricultural land also witnessed a marked reduction, signaling Kaduwela's transition from a predominantly agrarian landscape to a suburban-urban corridor within the Colombo metropolitan region. In 2002, agriculture was widespread, but by 2012, a large portion of farmland had been converted to residential and commercial zones, especially along key transportation routes. The 2024 map shows only isolated patches of farmland remaining on the municipal periphery, where urban development pressure is relatively low. This transformation raises food security concerns and underscores the importance of land-use policies aimed at preserving remaining agricultural zones and promoting urban agriculture initiatives.

In contrast, in Figure 11, built-up areas expanded markedly and became the dominant land-use category by 2024. From being concentrated in the southwest and central areas in 2002, built-up zones extended along major roads by 2012, forming linear urban patterns. By 2024, urban development had become more dispersed and fragmented, spreading into semi-urban and peri-urban areas. This expansion reflects both core densification and outward sprawl, creating challenges for infrastructure, transportation, drainage, and environmental management.

Overall, the LULC results confirm a clear transition from agricultural and vegetated landscapes toward increasingly built-up environments, with expansion occurring in both central and peripheral areas. These LULC changes have important implications for municipal governance and sustainability. Rapid built-up growth requires stronger spatial planning, zoning enforcement, and infrastructure provision to prevent unregulated expansion. The loss of forests and wetlands underscores the need to integrate environmental conservation and green infrastructure into urban policy.

Declining agricultural land also highlights the importance of protecting remaining productive land where appropriate. Together, these changes show that Kaduwela’s integration into the Colombo metropolitan region has created both development opportunities and environmental management challenges.

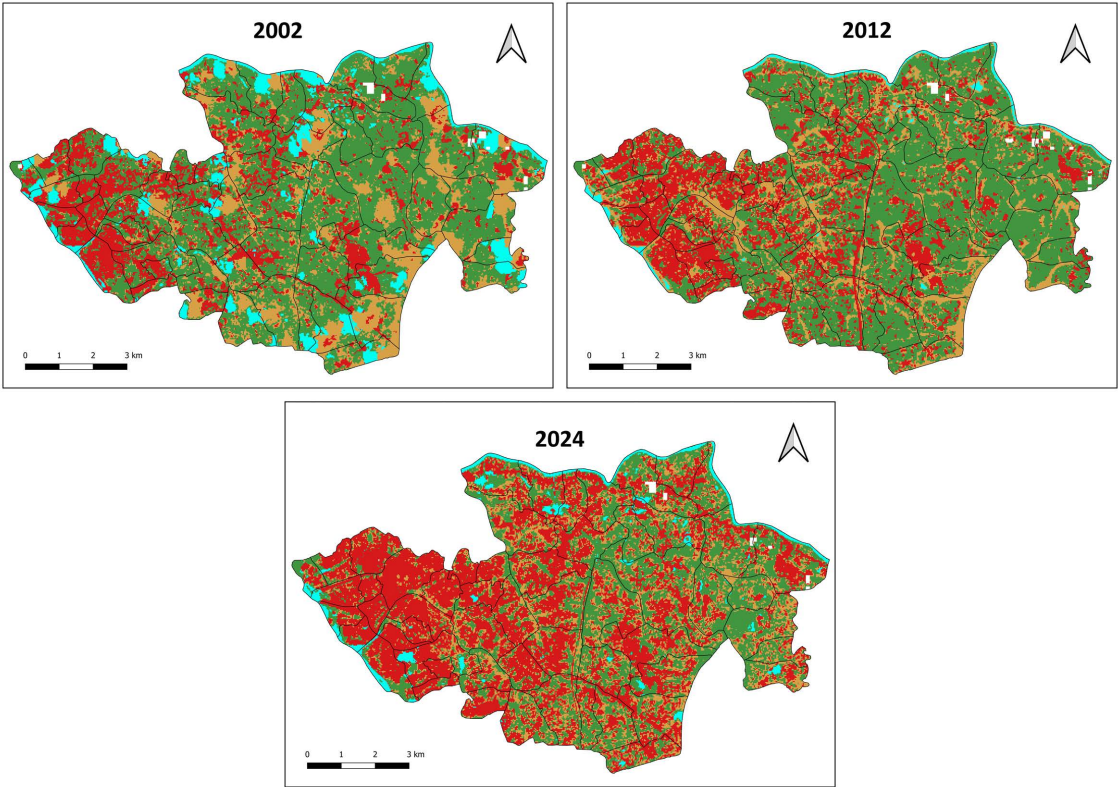


Figure 6. Land-use and land-cover changes in the Kaduwela Municipal Council (KMC) area in 2002, 2012, and 2024
 Note: Cyan indicates water bodies, green indicates vegetation cover, yellow-brown indicates agricultural land, and red indicates built-up land.

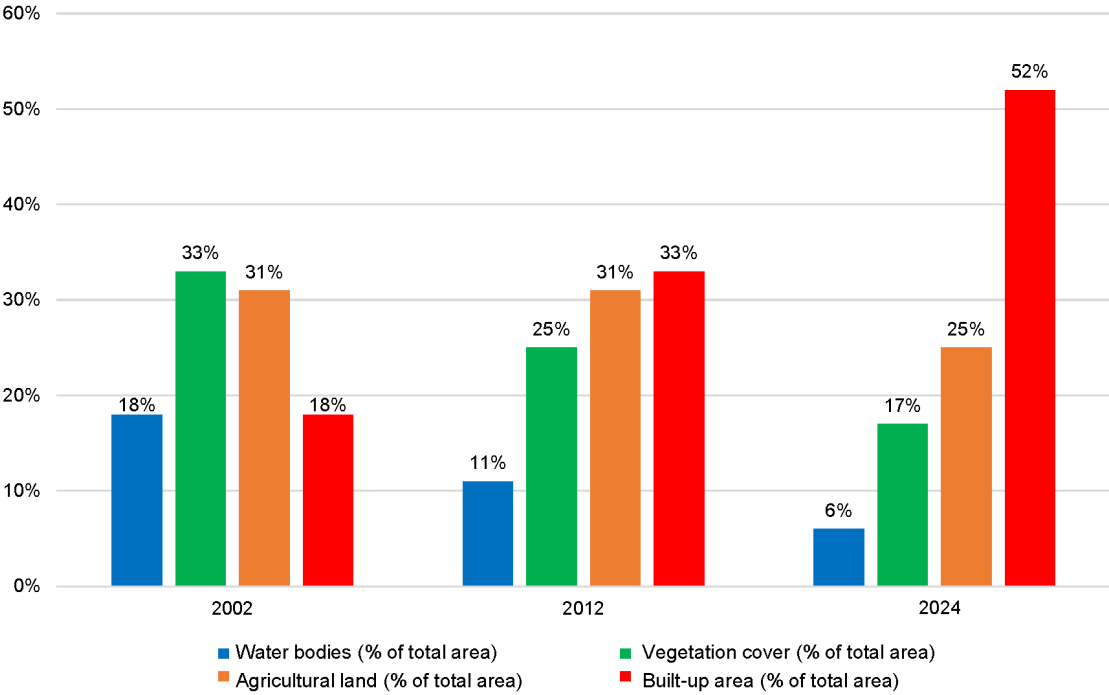


Figure 7. Percentage distribution of land-use and land-cover categories in the Kaduwela Municipal Council (KMC) area in 2002, 2012, and 2024

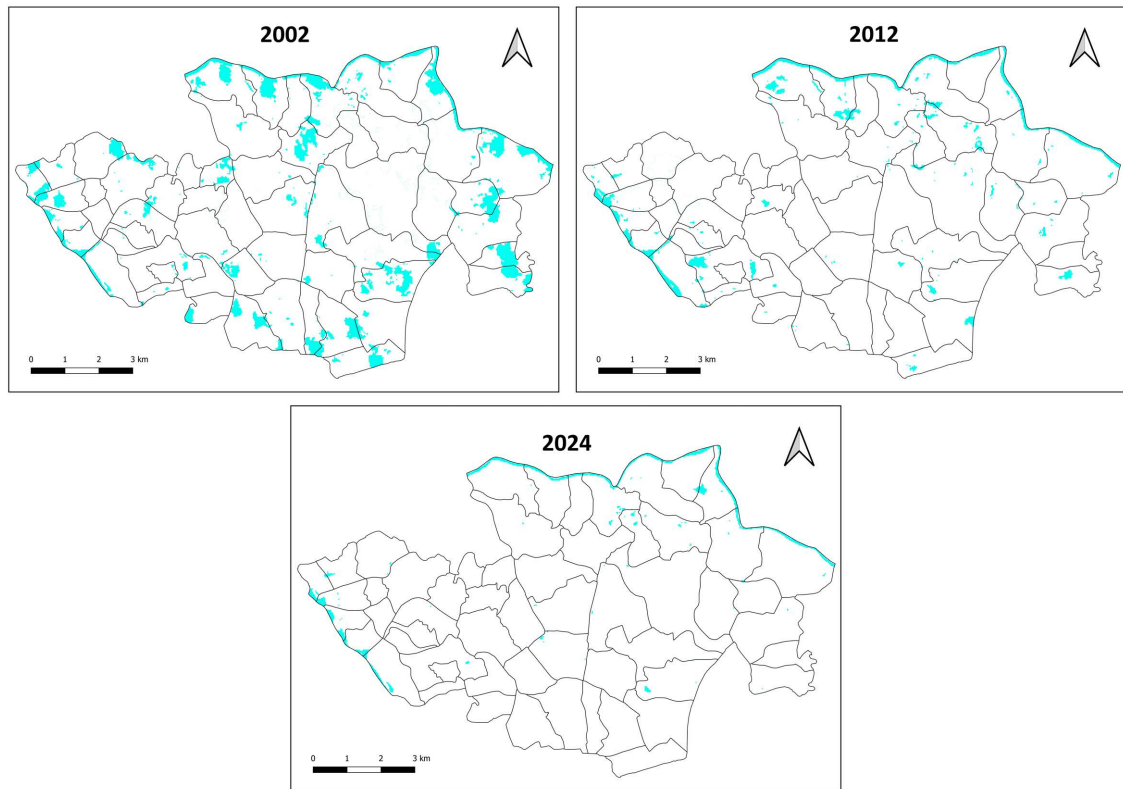


Figure 8. Decline in water bodies in the Kaduwela Municipal Council (KMC) area in 2002, 2012, and 2024
 Note: The cyan areas indicate water bodies.

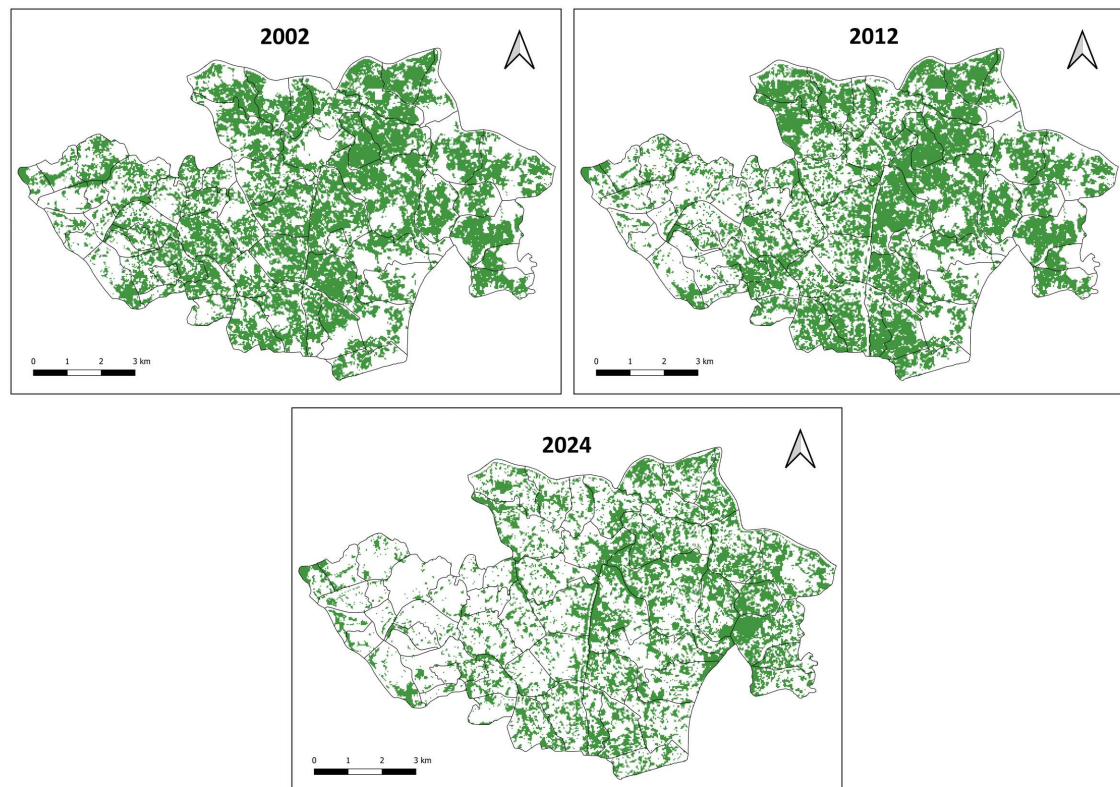


Figure 9. Decline in vegetation cover in the Kaduwela Municipal Council (KMC) area in 2002, 2012, and 2024
 Note: The green areas indicate vegetation cover.

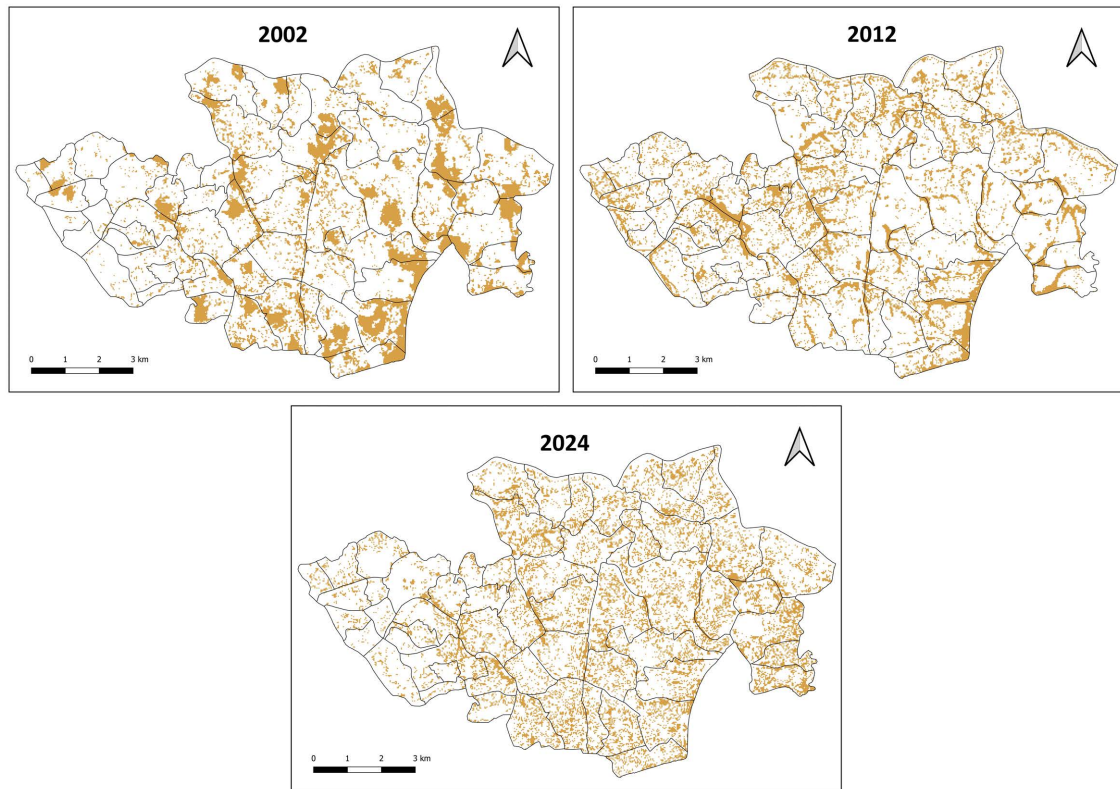


Figure 10. Decline in agricultural land in Kaduwela Municipal Council (KMC) in 2002, 2012, and 2024
 Note: The yellow-brown areas indicate agricultural land.

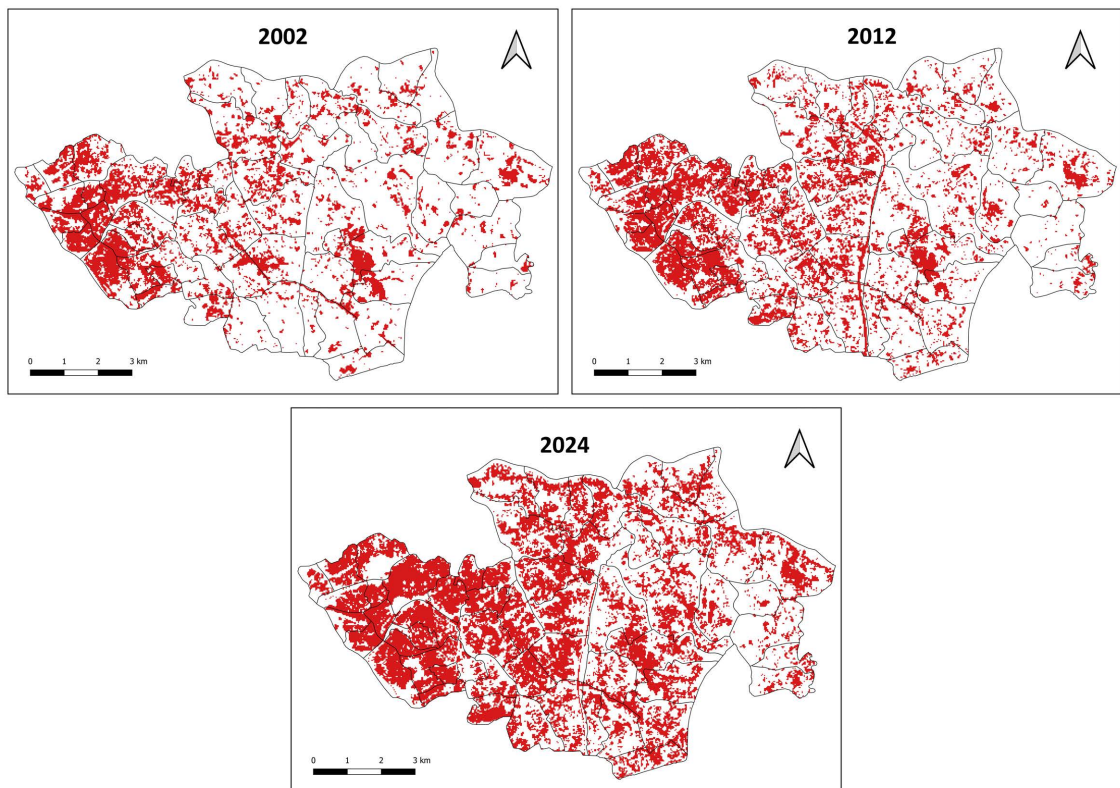


Figure 11. Expansion of built-up areas in Kaduwela Municipal Council (KMC) in 2002, 2012, and 2024
 Note: The red areas indicate built-up area.

4.3 Determinants of Urban Sprawl: Regression Results from Kaduwela Municipal Council

The regression analysis of the KMC area provides a quantitative understanding of the relationship between various demographic, infrastructural, and socioeconomic factors and the USI. The results reveal complex and context-specific dynamics that diverge from conventional urban economic expectations, reflecting the unique urbanization trajectory of Kaduwela within the Colombo metropolitan region.

4.3.1 Overall model performance of the regression analysis

The overall performance of the regression model is summarized in Table 1. The regression produced a multiple R value of 0.538, indicating a moderate correlation between the predicted and observed values of urban sprawl. This suggests that while the selected predictors are collectively meaningful, the relationship is not particularly strong, pointing to the multifaceted nature of urban sprawl. The R^2 value of 0.290 shows that approximately 29% of the variance in the USI is explained by the independent variables, while the remaining 71% is influenced by other factors not captured in the model. Although this may seem modest, such explanatory power is typical in urban studies, where spatial and socioeconomic processes are shaped by numerous interrelated forces, including land market behavior, informal housing dynamics, environmental constraints, and local governance decisions.

The adjusted R^2 value of 0.154 indicates that about 15% of the variation in sprawl is explained when model complexity is considered. The standard error of 7.16×10^{-5} suggests relatively small residuals, but this should be interpreted together with the limited adjusted R^2 value.

The low R^2 value is justified by the inherent complexity of urban systems in developing contexts like Sri Lanka, where urban sprawl results from overlapping spatial, political, and informal processes. Factors such as irregular land subdivision, unregulated construction, socio-political land allocation, and informal settlements are difficult to quantify and thus excluded from the regression model. Moreover, the rapid pace of post-2000 urbanization in Kaduwela, coupled with limited availability of high-resolution spatial data, likely contributes to statistical underrepresentation of certain dynamics. Therefore, while the model captures key measurable determinants, it does not encompass the full spectrum of underlying causes shaping sprawl.

Table 1. Model summary for the multiple regression analysis of the Urban Sprawl Index (USI)

Regression Statistic	Value
Multiple correlation coefficient (R)	0.538
Coefficient of determination (R^2)	0.290
Adjusted coefficient of determination (adjusted R^2)	0.154
Standard error of the estimate	7.16×10^{-5}
Number of observations (N)	57
F -statistic	2.129
Regression degrees of freedom (df_{reg})	9
Residual degrees of freedom (df_{res})	47
Overall model significance (p -value)	0.045 ($p < 0.05$)

4.3.2 Regression coefficients and key determinants of urban sprawl

As shown in Table 2, the regression results indicate that population growth rate was the only statistically significant determinant of the USI ($\beta = -1.09 \times 10^{-6}$, $p = 0.003$), with a negative effect on urban sprawl. This result suggests that rapid population growth in Kaduwela has been accompanied by vertical densification rather than horizontal expansion, possibly due to land scarcity, zoning regulations, and municipal policies encouraging infill and compact development.

Population density ($\beta = -1.40 \times 10^{-8}$, $p = 0.158$) also exhibited a negative relationship with sprawl but lacked statistical significance. This implies that the relationship between density and sprawl is not statistically confirmed in the model. Kaduwela may demonstrate both inner-area densification and peripheral low-density development, but this should be interpreted together with the spatial LULC results.

In-migration rate ($\beta = 1.03 \times 10^{-6}$, $p = 0.553$) showed a positive but statistically insignificant association with sprawl. Therefore, migration may contribute to urbanization, but the present model does not provide strong statistical evidence that it directly drives spatial expansion. Similarly, household density ($\beta = -4.81 \times 10^{-6}$, $p = 0.484$) showed an insignificant negative relationship, indicating that household clustering alone does not substantially explain the spatial spread of development in this model.

Infrastructure variables such as road density ($\beta = -2.39 \times 10^{-7}$, $p = 0.569$) and vehicles per km^2 ($\beta = 3.87 \times 10^{-8}$, $p = 0.339$) demonstrated weak relationships with sprawl, both statistically insignificant. Although road networks and vehicle ownership are often expected to influence suburban expansion, the present results do not confirm

a significant effect in the KMC area. This may suggest that additional factors, such as land-use regulation, congestion, transport accessibility, or spatial constraints, need to be considered in future models.

The employment rate ($\beta = 1.96 \times 10^{-7}$, $p = 0.393$) and sectoral population variables for urban ($\beta = 6.02 \times 10^{-9}$, $p = 0.278$) and semi-urban populations ($\beta = 2.99 \times 10^{-9}$, $p = 0.641$) were also positive but statistically insignificant. These results suggest that employment and sectoral population distribution may be related to urban development patterns, but their effects are not statistically supported in the present regression model.

Table 2. Regression coefficients for the determinants of the Urban Sprawl Index (USI)

Variable	Coefficients (β)	Standard Error	t-Statistic	p-Value
Intercept-USI	3.02×10^{-4}	4.74×10^{-5}	6.381	<0.001
Population density	-1.40×10^{-8}	9.75×10^{-9}	-1.434	0.158
Population growth rate (2002–2023)	-1.09×10^{-6}	3.53×10^{-7}	-3.086	0.003
In-migration rate	1.03×10^{-6}	1.72×10^{-6}	0.598	0.553
Household density	-4.81×10^{-6}	6.81×10^{-6}	-0.706	0.484
Road density	-2.39×10^{-7}	4.17×10^{-7}	-0.573	0.569
Vehicles per km ²	3.87×10^{-8}	4.01×10^{-8}	0.966	0.339
Employment rate	1.96×10^{-7}	2.27×10^{-7}	0.862	0.393
Urban sector population	6.02×10^{-9}	5.48×10^{-9}	1.097	0.278
Semi-urban sector population	2.99×10^{-9}	6.37×10^{-9}	0.469	0.641

4.4 Planning and Governance Implications

The regression results indicate three main patterns. Since population growth rate is the only statistically significant predictor, the interpretation focuses primarily on this variable, while the remaining variables are discussed as weak or non-significant associations.

First, population growth may be associated with compact growth or densification. The negative and significant association between population growth and sprawl suggests that recent population growth in Kaduwela may have been partly absorbed through compact development or densification rather than extensive outward expansion. However, this interpretation requires further evidence, such as building-height data, housing-density data, floor-area records, or building permit information. Therefore, it should be regarded as a possible explanation rather than a confirmed conclusion.

Second, infrastructure variables show weak statistical influence. Road density and vehicle density do not significantly explain sprawl in the present model. This may indicate that transport development alone does not sufficiently account for land-use change in Kaduwela. Other factors, such as zoning enforcement, land availability, land-market behavior, and environmental constraints, may also play important roles.

Third, migration and sectoral population variables have limited explanatory effects. In-migration rate and sectoral population variables show weak and statistically insignificant relationships with the USI. This suggests that these variables are not dominant measurable drivers of urban sprawl in the current model, although they may still influence urban transformation indirectly.

Based on these findings, several planning and governance implications can be identified:

- **Support compact and well-regulated growth:** The negative association between population growth and sprawl suggests the potential importance of compact development. However, more evidence is needed before concluding that existing densification policies have been effective. Future planning should therefore encourage well-regulated infill development, mixed-use zoning, and appropriate redevelopment of underutilized land.
- **Strengthen integrated land-use and infrastructure planning:** Since road density and vehicle density were not statistically significant, infrastructure planning should be more closely integrated with land-use regulation, zoning control, and environmental management. This would help reduce the risk of fragmented or corridor-based expansion as infrastructure networks continue to develop.
- **Enhance data and modeling capacity:** The modest explanatory power of the regression model indicates the need for more comprehensive data. Future studies should incorporate additional spatial and governance-related variables, such as land price trends, zoning enforcement, building density, building permits, informal settlement patterns, and environmental constraints.
- **Protect environmental assets:** The LULC results show a decline in water bodies, vegetation, and agricultural land. Therefore, wetland protection, flood-risk management, green-space conservation, and agricultural land protection should be considered as important components of future municipal planning.

Overall, the regression results suggest that urban sprawl in Kaduwela is only moderately explained by the selected demographic and infrastructural variables. Population growth was the only statistically significant determinant, while

other variables showed weak or non-significant associations. The relatively low R^2 and adjusted R^2 values indicate that other factors, such as land-use regulation, spatial constraints, informal development, land-market behavior, and environmental conditions, may also influence urban expansion. Therefore, more integrated planning, improved data collection, and stronger municipal governance are needed to manage future urban growth in the KMC area.

5 Conclusions and Recommendations

5.1 Conclusions

The findings from the population, land-use, and regression analyses indicate that KMC is undergoing a transitional phase of urbanization characterized by both population concentration and peripheral expansion. Based on the latest available demographic data up to 2023, the municipality experienced continued population growth, while the LULC results for 2002–2024 show substantial spatial transformation, particularly the decline of agricultural land, vegetation cover, and water bodies, together with the expansion of built-up land. These demographic and spatial changes highlight a dual urban challenge: accommodating development pressure while maintaining environmental sustainability and spatial equity.

The regression analysis suggests that urban sprawl in Kaduwela is only moderately explained by the selected demographic, infrastructural, and socioeconomic variables. Population growth emerged as the only statistically significant determinant. Its negative relationship with the USI may indicate that part of the recent population growth has been absorbed through compact development or densification rather than extensive outward expansion. However, this interpretation should be treated cautiously, as further evidence, such as building-height data, housing-density records, floor-area data, or building permit information, would be required to confirm the extent of densification.

The limited explanatory power of the regression model also indicates that other factors may influence urban expansion in Kaduwela. These may include land-use regulation, zoning enforcement, informal development, land-market behavior, environmental constraints, and local governance capacity. Therefore, urban sprawl in the KMC area should be understood as a complex process shaped by both measurable demographic and infrastructural variables and broader institutional and spatial conditions.

Overall, the study identifies a mixed urban form in Kaduwela, where compact development in some areas coexists with fragmented and low-density expansion in peri-urban zones. This pattern requires an integrated and adaptive governance approach that links land-use planning, infrastructure provision, environmental protection, and municipal coordination.

From a policy implementation perspective, a phased action framework may support more sustainable urban management in Kaduwela. In the short term (1–3 years), the Urban Development Authority and KMC should prioritize strengthening land-use regulation, updating zoning plans, and improving GIS-based monitoring systems to track land-use change and identify areas under development pressure.

In the medium term (3–7 years), coordinated efforts between the Municipal Council, Western Provincial Council, and relevant national agencies should focus on improving infrastructure capacity, strengthening public transport integration, and guiding compact urban development through better land-use and transport planning.

In the long term (7–15 years), strategic attention should be given to building a more resilient and sustainable urban structure. This may include wetland protection, ecosystem restoration, conservation of remaining agricultural and green areas, and the development of a more balanced spatial growth model to reduce excessive pressure on the Colombo metropolitan core. These measures should be supported by improved institutional coordination and, where appropriate, community and private-sector participation.

In conclusion, Kaduwela's urban transformation presents both opportunities and challenges. While compact growth may support more efficient land use, uncontrolled peripheral expansion continues to pose risks to long-term sustainability. Strengthening spatial planning, environmental governance, data collection, and inter-agency coordination will be essential for guiding Kaduwela toward a more balanced, resilient, and sustainable urban future.

5.2 Recommendations for Future Studies

While this study provides useful insights into urban sprawl in the KMC area, it also highlights several areas for future research. First, future studies should incorporate longitudinal household-level data to better understand how migration, income levels, housing preferences, and household characteristics influence urban sprawl. Such micro-level analyses would complement the municipal-scale approach used in this study.

Second, future research should consider more advanced spatial modeling techniques, such as cellular automata, Markov chain models, or agent-based simulations, to predict future land-use patterns under different planning and policy scenarios. These approaches would help municipalities assess the possible long-term effects of infrastructure development, zoning decisions, and environmental protection measures.

Third, future studies should include additional governance and spatial variables that were not fully captured in the present regression model. These may include zoning enforcement, land prices, building density, building permits,

informal settlement patterns, transport accessibility, and policy implementation capacity. Including such variables may improve the explanatory power of future models.

Fourth, given the environmental sensitivity of Kaduwela, future research should integrate climate-related factors into urban sprawl analysis. In particular, the relationship between land-use change, flood risk, wetland loss, and urban heat island effects should be examined to support climate-resilient urban planning.

Finally, comparative studies across other peri-urban municipalities in Sri Lanka would be valuable. Such studies could help identify whether Kaduwela's urbanization pattern is unique or reflects broader trends in Sri Lankan municipal development. This would provide a stronger basis for national-level urban planning and policy formulation.

In summary, future research should adopt more data-rich, spatially detailed, and interdisciplinary approaches to deepen the understanding of urban sprawl dynamics. This would support policymakers and planners in developing adaptive strategies for sustainable urban growth in Kaduwela and other rapidly urbanizing municipalities in Sri Lanka.

Data Availability

The data used to support the research findings are available from the corresponding author upon request.

Conflicts of Interest

The author declares no conflicts of interest.

Declaration on the Use of Generative AI and AI-assisted Technologies

The author declares that generative artificial intelligence (AI) or AI-assisted technologies were used only for language editing, grammar correction, and academic polishing of the manuscript. The authors reviewed and edited all AI-assisted text and remain fully responsible for the accuracy, originality, and integrity of the manuscript.

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